2 The relational model: ER modelling

Week 2 · Worksheet

Name _____ Date _____

Exercises

Show your work. Each exercise has a hint at the end: try without it first.

1. Tree or not? Warm-up · Lesson 1

A streaming service has songs and playlists. One song can be in many playlists, and a playlist holds many songs. Why is this awkward in a hierarchical model?

Hint: In a tree, how many parents can a record have?

2. Explain data independence Practice · Lesson 1

The streaming service moves its song table to faster disks and adds an index on title. Which programs must change in a relational system, and why?

Hint: Do the queries mention disks or indexes?

3. Name the parts Warm-up · Lesson 2

STATION(station_id, name, capacity, latitude, longitude) has 48 stations. Give its degree, its cardinality, and a sensible domain for capacity.

Hint: Count columns, then rows. Can a station hold -3 bikes?

4. Is it a valid relation? Practice · Lesson 2

A table LISTENER has two identical rows (7, "mia@example.com"), and a column devices holding "phone; laptop". Which relational rules are broken?

Hint: A relation is a set, and cells are atomic.

5. Superkey or candidate key? Warm-up · Lesson 3

ALBUM(album_id, upc_code, title, release_year), where album_id and upc_code are each unique. Classify: {album_id}, {upc_code}, {album_id, title}, {title}.

Hint: Is it unique? If so, can you drop an attribute and stay unique?

6. Pick the primary key Practice · Lesson 3

A vet clinic identifies pets by microchip number, but some rescued pets arrive without a chip and get one weeks later. Is microchip a good primary key?

Hint: Can a primary key be NULL, even for a while?

7. Find the composite key Challenge · Lesson 3

A school timetable stores SLOT(room_id, weekday, period, group_id, subject_code). A room can host one class per period. Which composite key follows from that rule? Is there another candidate key?

Hint: A class group also cannot be in two rooms at once.

8. Choose the delete rule Warm-up · Lesson 4

Pick RESTRICT, CASCADE, or SET NULL: (a) deleting a playlist and its playlist_song rows; (b) deleting an artist who still has songs; (c) a listener deletes their account, and their reviews should stay as "anonymous".

Hint: Do the children make sense without the parent?

9. Spot the violations Practice · Lesson 4

STATION has ids 1, 2, 3. Which inserts fail and why? (a) bike(10, 'City', 2) (b) bike(NULL, 'City', 1) (c) bike(11, 'Cargo', 7) (d) bike(12, 'City', NULL)

Hint: Check the primary key first, then the foreign key.

10. Name the cardinality Warm-up · Lesson 5

Bike sharing: (a) rider and membership card, one card per rider; (b) station and bike, a bike is docked at one station; (c) rider and bike over time, through trips.

Hint: Say "each … has …" in both directions.

11. Where does the attribute go? Practice · Lesson 5

A listener can follow many artists; an artist has many followers. You must store the date each follow started and whether notifications are on. Design the tables.

Hint: Do the date and the setting describe the listener, the artist, or the pair?

12. Degree and roles Warm-up · Lesson 6

Give the degree of each relationship: (a) a listener follows another listener; (b) a listener saves a song; (c) a teacher teaches a subject to a class group.

Hint: Count the different entity types involved, then check if one repeats.

13. Requirements to ER Practice · Lesson 6

Vet clinic: "Owners (id, name, phones) bring pets (id, name, species, birth date). Each pet has exactly one owner. Vets (id, name) carry out visits; each visit is one pet with one vet on a date, with a reason. Each pet's vaccine doses are numbered 1, 2, 3…" List entities, attributes with their types, relationships with (min, max), and any weak entity.

Hint: Phones repeat; a dose number repeats across pets.

14. Classify and map a hierarchy Practice · Lesson 7

A vet clinic stores staff (id, name, hire date). Vets also have a licence number; nurses have a shift. Some staff are receptionists with neither. Nobody is both vet and nurse. Classify the hierarchy and write the tables with option A.

Hint: Can someone be in no subtype? In two?

15. Draw the ends Warm-up · Lesson 8

Write crow's-foot ends for: "each album belongs to exactly one artist; each artist has zero or many albums".

Hint: The count of artists per album goes next to Artist.

16. Translate Chen to crow's foot Practice · Lesson 8

Chen: [GROUP] =1= <has tutor> —1— [TEACHER], with a double line on the Group side. Write it in crow's foot and say where the foreign key goes.

Hint: Double line = total participation = minimum 1.

17. Model a climbing gym Challenge · Lesson 9

"A climbing gym sells memberships to climbers (id, name, birth date, several emergency contacts). Each visit records the climber, the date, and the amount paid. Instructors (id, name) run courses (code, title, level); a climber can enrol in many courses and a course has many climbers, with the enrolment date. The owner wants the income of each month and each climber's age." Apply the five steps and list entities, attributes (with types), relationships with cardinalities, and the derived values.

Hint: Visits are events recorded one by one. Who runs a course: one instructor or many?

18. Map the bike-sharing model Practice · Lesson 10

ER: Station(station_id, name, address{street, city}); Bike(bike_id, model); Rider(rider_id, email, phones*); Station 1:N Bike (bike may be out riding); Rider N:M Bike through "rides" with started_at and ended_at; Dock is weak, owned by Station, partial key dock_no. Write the relations.

Hint: Follow the seven steps in order; * means multivalued; {…} means composite.

19. A logical model from a real scenario Challenge · Lesson 10

A school timetable: teachers (id, name, emails*); subjects (code, title, weekly hours); class groups (id, label) each with exactly one tutor teacher, and a teacher tutors at most one group; rooms (id, seats). Each timetable slot places one group, one subject, one teacher, and one room on a weekday and period. Build the full logical model with keys, UNIQUE, and NOT NULL.

Hint: The slot is a relationship of degree four. Which pairs can never repeat in the same period?

20. Pick the types Warm-up · Lesson 11

Choose MySQL and PostgreSQL types for: a rider id generated automatically; a subscription price like 12.90; the moment a trip started, for riders in several time zones; whether a bike needs repair.

Hint: Money is exact; time zones need the TZ variant.

21. Translate a table Practice · Lesson 11

Rewrite for PostgreSQL: CREATE TABLE album (album_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(100) NOT NULL, released YEAR, is_single TINYINT(1) DEFAULT 0, artist_id INT, CONSTRAINT fk_album_artist FOREIGN KEY (artist_id) REFERENCES artist(artist_id) ON DELETE RESTRICT);

Hint: Three types change; the constraint stays the same.

22. Write the four statements Warm-up · Lesson 12

Using station(station_id, name, capacity): add stations 5 "Plaza Norte" (20) and 6 "Muelle" (12) in one statement; list names and capacities from largest to smallest as "station" and "docks"; raise Muelle to 16 docks; remove station 6.

Hint: Multi-row VALUES, AS aliases, ORDER BY … DESC, WHERE on the key.

23. Why did the DELETE fail? Practice · Lesson 12

DELETE FROM station WHERE station_id = 5; fails with a foreign key error. Bikes reference station 5 through a constraint with ON DELETE RESTRICT. Give two correct ways to close the station.

Hint: Move the children first, or change the rule.

Quick check

Circle one answer.

24. What was the main weakness of hierarchical and network databases?

A) Programs had to navigate stored paths, so new questions and structure changes meant new code

B) They could not store text C) They were always slower than relational systems

25. LISTENER(listener_id, email, display_name). listener_id and email are each unique. Which set is a superkey but NOT a candidate key?

A) {email, display_name} B) {email} C) {display_name}

26. A streaming service numbers tracks 1, 2, 3… inside each album. TRACK(album_id, track_no, song_id). What is the primary key?

A) (album_id, track_no) B) track_no C) album_id

27. A playlist is deleted. Its rows in playlist_song should disappear with it. Which rule fits playlist_song.playlist_id?

A) ON DELETE CASCADE B) ON DELETE RESTRICT C) ON DELETE SET NULL

28. Each pet has exactly one owner; an owner may have several pets. Where does the link go?

A) pet.owner_id, NOT NULL B) owner.pet_id C) A junction table owner_pet

29. "A vet prescribes a drug to a pet, with a dosage." What is the degree of this relationship?

A) 3 (ternary) B) 2 (binary) C) 1 (unary)

30. In crow's-foot notation, ARTIST ||——o< ALBUM means:

A) Each artist has zero or many albums; each album has exactly one artist

B) Each album has zero or many artists; each artist has exactly one album

C) Artists and albums are N:M

31. Students join clubs (N:M) and each membership stores a role, like "captain". After mapping, where is role?

A) In the junction table membership(student_id, club_id, role) B) In student C) In club

Answer key

1. A tree gives each record one parent. A song would need to be copied under every playlist, so its title would be stored many times and could drift out of sync. The relational model stores the song once and lists the pairs (playlist, song) in a junction table.
2. None. Queries name tables and columns, not storage. The optimiser simply starts using the index. In a navigational system, code that walked the old access path might have to be rewritten.
3. Degree 5, cardinality 48. Capacity: whole numbers from 1 up to a realistic maximum, for example 1–60.
4. Duplicate tuples are not allowed in a set, so one row must go (a primary key would prevent it). devices holds a list, breaking atomicity: move devices to a table LISTENER_DEVICE(listener_id, device).
5. {album_id} and {upc_code}: candidate keys (and superkeys). {album_id, title}: superkey, not minimal. {title}: not a key, two albums can share a title.
6. No. A primary key can never be NULL. Use a surrogate pet_id as the primary key and keep microchip as an alternate key declared UNIQUE, allowing NULL until the chip is placed.
7. {room_id, weekday, period} is a candidate key. Because a group is in one place per period, {group_id, weekday, period} is another. Pick one as primary (or add a surrogate slot_id) and declare the other UNIQUE.
8. (a) CASCADE: entries mean nothing without their playlist. (b) RESTRICT: removing an artist with songs is probably a mistake. (c) SET NULL on review.listener_id, which must therefore allow NULL.
9. (a) OK. (b) breaks entity integrity: a primary key cannot be NULL (SQLite would generate an id for an INTEGER PRIMARY KEY, but the rule stands). (c) breaks referential integrity: station 7 does not exist. (d) OK: station_id allows NULL, the bike is out riding.
10. (a) 1:1: card.rider_id UNIQUE. (b) 1:N: bike.station_id. (c) N:M: trip is the junction table (rider_id, bike_id) with its own attributes started_at and ended_at.
11. listener(listener_id PK, …), artist(artist_id PK, …), follow(listener_id FK, artist_id FK, since, notify, PK(listener_id, artist_id)). Both attributes belong to the pair.
12. (a) unary (recursive), with roles follower and followed. (b) binary. (c) ternary: teacher, subject, and class group.
13. Owner(owner_id key, name, phones multivalued). Pet(pet_id key, name, species, birth_date; age would be derived). Vet(vet_id key, name). Visit(visit_id key, date, reason), or an N:M "visits" between Pet and Vet with date and reason. Dose: weak entity owned by Pet, partial key dose_no. Pet (1,1) — owns — (0,N) Owner; Pet (0,N) — visit — (0,N) Vet; Dose (1,1) — of — (0,N) Pet.
14. Disjoint and partial. staff(staff_id PK, name, hired_on); vet(staff_id PK FK -> staff, licence_no UNIQUE); nurse(staff_id PK FK -> staff, shift).
15. ARTIST ||——o< ALBUM. Next to Artist: || (exactly one). Next to Album: o< (zero or many).
16. GROUP o|——|| TEACHER: each group has exactly one tutor (|| next to Teacher); each teacher tutors zero or one group (o| next to Group). Put tutor_id in class_group as NOT NULL UNIQUE, the side with total participation.
17. Entities: CLIMBER(climber_id key, name, birth_date, emergency_contacts multivalued), VISIT(visit_id key, visit_date, amount), INSTRUCTOR(instructor_id key, name), COURSE(code key, title, level). Relationships: CLIMBER (1,1)–(0,N) VISIT; INSTRUCTOR (1,1)–(0,N) COURSE (assuming one instructor per course; ask the owner); CLIMBER N:M COURSE "enrols" with enrolled_on. Derived: monthly income = SUM(amount) of the month's visits; age from birth_date. Neither is stored.
18. station(station_id PK, name, street, city); dock(station_id FK, dock_no, PK(station_id, dock_no)); bike(bike_id PK, model, station_id FK NULL); rider(rider_id PK, email UNIQUE); rider_phone(rider_id FK, phone, PK(rider_id, phone)); trip(rider_id FK, bike_id FK, started_at, ended_at, PK(rider_id, bike_id, started_at)) or with a surrogate trip_id.

19. teacher(teacher_id PK, name); teacher_email(teacher_id FK, email, PK(teacher_id, email)); subject(subject_code PK, title, weekly_hours); class_group(group_id PK, label, tutor_id FK NOT NULL UNIQUE); room(room_id PK, seats); slot(slot_id PK, group_id FK, subject_code FK, teacher_id FK, room_id FK, weekday, period, all NOT NULL, UNIQUE(group_id, weekday, period), UNIQUE(room_id, weekday, period), UNIQUE(teacher_id, weekday, period)).
20. Id: INT AUTO_INCREMENT / SERIAL. Price: DECIMAL(6,2) / NUMERIC(6,2). Start: DATETIME (store UTC by convention) / TIMESTAMPTZ. Repair flag: TINYINT(1) or BOOLEAN / BOOLEAN.
21. CREATE TABLE album (album_id SERIAL PRIMARY KEY, title VARCHAR(100) NOT NULL, released SMALLINT, is_single BOOLEAN DEFAULT FALSE, artist_id INT, CONSTRAINT fk_album_artist FOREIGN KEY (artist_id) REFERENCES artist(artist_id) ON DELETE RESTRICT);
22. INSERT INTO station (station_id, name, capacity) VALUES (5, 'Plaza Norte', 20), (6, 'Muelle', 12); SELECT name AS station, capacity AS docks FROM station ORDER BY capacity DESC; UPDATE station SET capacity = 16 WHERE station_id = 6; DELETE FROM station WHERE station_id = 6;
23. (1) UPDATE bike SET station_id = 4 WHERE station_id = 5; then DELETE the station. (2) Redefine the constraint with ON DELETE SET NULL, so the bikes stay with no station. Even better for history: keep the row and mark it closed (is_open = FALSE).
24. **A** — Navigational access tied programs to the physical paths. Ironically they were often faster at first; their problem was flexibility.
25. **A** — {email, display_name} is unique because email is, but it is not minimal. {email} is a candidate key; {display_name} is not unique at all.
26. **A** — Track 1 exists in every album, and an album has many tracks: only the pair is unique. A composite key.
27. **A** — The entries have no meaning without their playlist, so CASCADE. SET NULL is impossible anyway: playlist_id is part of the junction's primary key.
28. **A** — 1:N: the foreign key lives on the "many" side, and NOT NULL because every pet must have an owner. A junction table would allow several owners per pet.
29. **A** — Three entity types take part in one fact: vet, drug, pet. Dosage is an attribute of the relationship.
30. **A** — Read each end from the far entity: o< next to Album counts albums per artist; || next to Artist counts artists per album.
31. **A** — The role belongs to the pair: the same student can be captain of one club and a regular member of another.